Bitwuzla-BV_Parti at SMT-COMP 2026

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1 Introduction

Bitwuzla-BV_Parti is a wrapper SMT solver based on Bitwuzla [1] (v0.9.1). It participates in the QF_BV logic, with emphasis on the Parallel track. The solver keeps Bitwuzla as the base solver for ordinary SMT-LIB tasks and adds a lightweight divide-and-conquer layer for generating and scheduling BV subproblems.

BV_Parti is a continuation of our work on dynamic variable-level partitioning for SMT solving, first studied in AriParti [2]. While AriParti focused on arithmetic theories, BV_Parti explores how the same distributed partitioning idea can be adapted to fixed-size bit-vectors. The full technical description of the current BV_Parti method is not yet published; a detailed paper will follow after the SMT-COMP submission.

2 System Description

BV_Parti consists of three main components: a master, a partitioner, and base solver workers.

1. The master is implemented in Python and manages task generation, scheduling, timeouts, and result aggregation.
2. The partitioner is implemented by extending the Bitwuzla front-end. It simplifies the current task and emits two standard SMT-LIB child tasks when a useful partition is found.
3. The base solver workers run Bitwuzla on generated subtasks. The framework only requires an executable solver that accepts SMT-LIB input and returns a standard SMT result.

The master maintains a tree of subtasks. If one child is satisfiable, the original task is satisfiable. If all children of a sound split are unsatisfiable, the parent is marked unsatisfiable. Failed or timed-out tasks are handled conservatively.

3 Features

Compared with plain Bitwuzla, Bitwuzla-BV_Parti adds the following features.

Dynamic parallel framework. The runtime keeps a pool of generated tasks and dispatches them to available cores. Hard tasks can be returned to the partitioner and split further, so the partition granularity is adjusted during solving rather than fixed in advance.

BV-aware semantic partitioning. The partitioner uses lightweight bit-vector information to select split guards. In the current implementation, this information is derived from feasible-domain summaries such as fixed bits and modular intervals. These summaries help generate child tasks that expose more word-level structure to Bitwuzla's normal preprocessing and bit-blasting pipeline.

Worker interface. Although the SMT-COMP configuration uses Bitwuzla 0.8.0 as the base solver, the wrapper communicates with workers through standard SMT-LIB files. This keeps the distributed framework separate from the sequential solver.

4 Relation to SPFD

BV_Parti and SPFD share the idea of using feasible-domain information over bit-vector terms. Their roles in SMT-COMP are different. SPFD is a derived solver that strengthens a single Bitwuzla run by preprocessing, while BV_Parti is a wrapper solver that uses similar information to guide parallel partitioning and scheduling.

5 Acknowledgements

We thank the authors and developers of Bitwuzla for their foundational work. Our tool builds on Bitwuzla, whose clean and well-structured codebase made our implementation and optimizations substantially easier. We are also grateful for Bitwuzla's high-performance solving infrastructure, which provides a strong basis for the effectiveness of our tool.

We further thank the developers of STP. Its existing implementation offered valuable experience and practical insights that helped guide the development of our work.

References

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